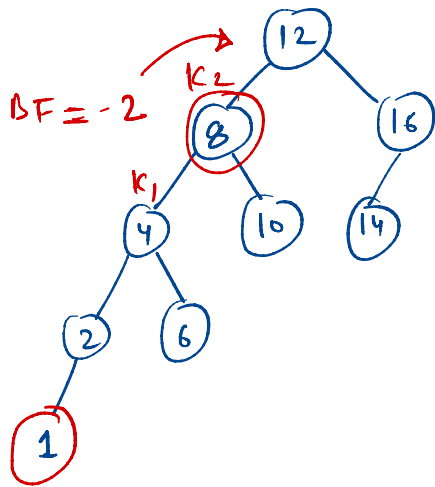
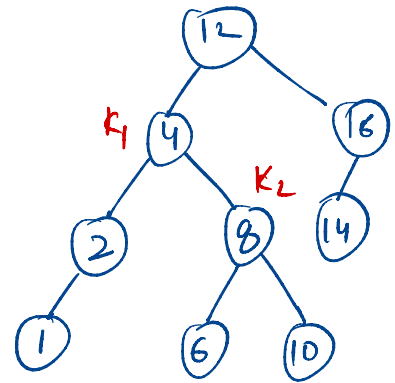


AVL Trees - III

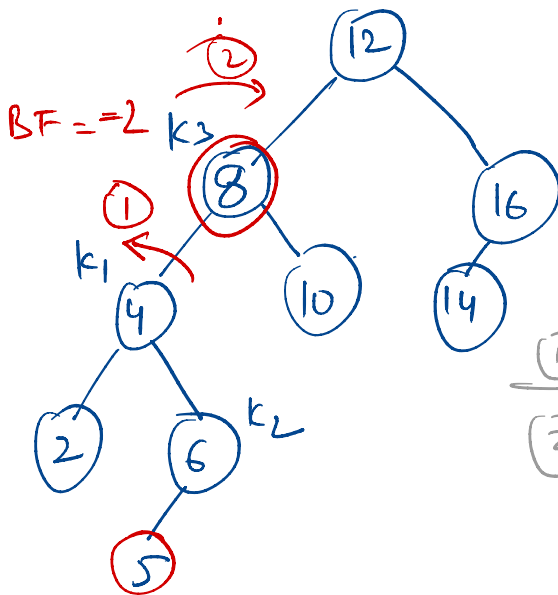
Examples : Case LL (right rotation)



right-rotate(k_2)

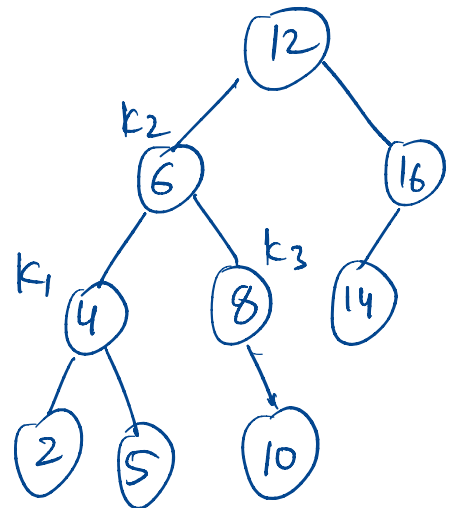


Case LR (left-right double rotation)



① left-rotate(k_1)

② right-rotate(k_3)



Implementing Add in an AVL Tree

- Perform normal BST add. When the recursive call returns, balance the node if there is an imbalance
↳ i.e, if BF becomes $-2/+2$.

- logic to detect the case for resolution:

- Current node BF $< -1 \Rightarrow$ Case LL or LR
↳ look at current node's left child BF

fix with R rotation | left child BF $< 0 \Rightarrow$ Case LL
fix with LR rotation | left child BF $> 0 \Rightarrow$ Case LR

- Current node BF $> 1 \Rightarrow$ Case RL or RR
↳ look at current node's right child BF

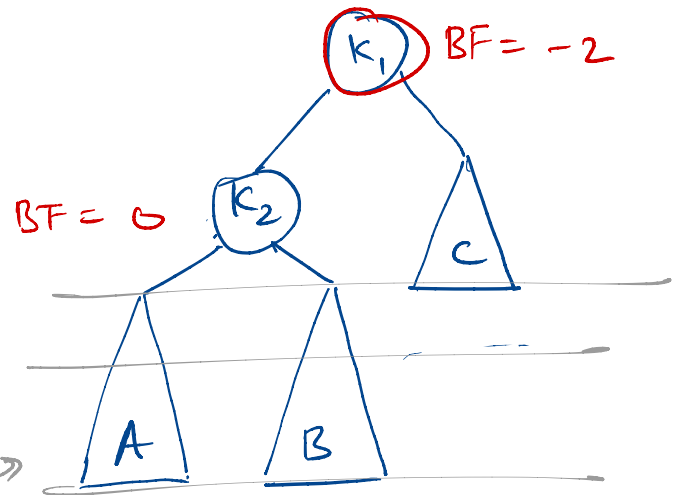
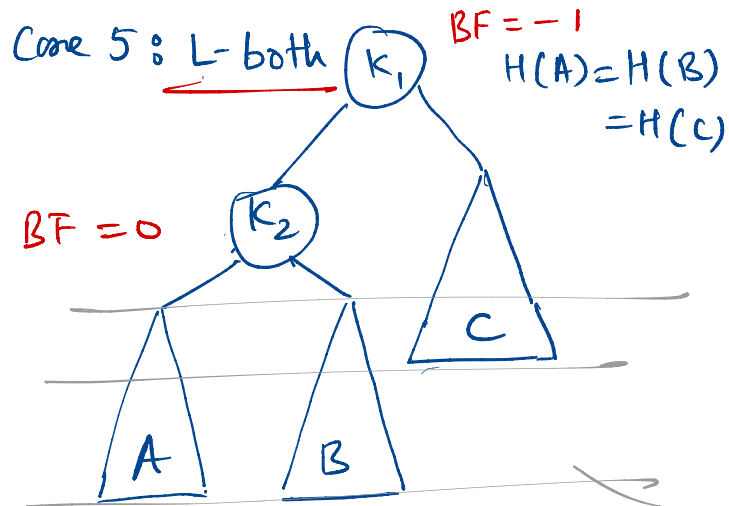
fix with RL rotation | Right child BF $< 0 \Rightarrow$ Case RL
fix with R rotation | Right child BF $> 0 \Rightarrow$ Case RR

AVL remove cases:

→ Similar cases as with adding: LL, LR, RL, & RR

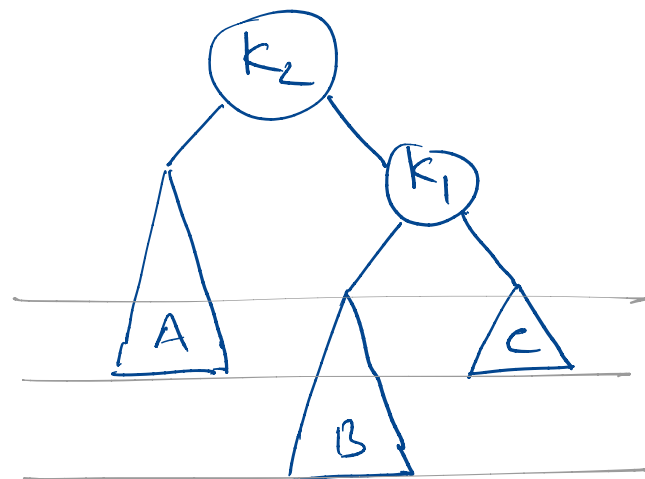
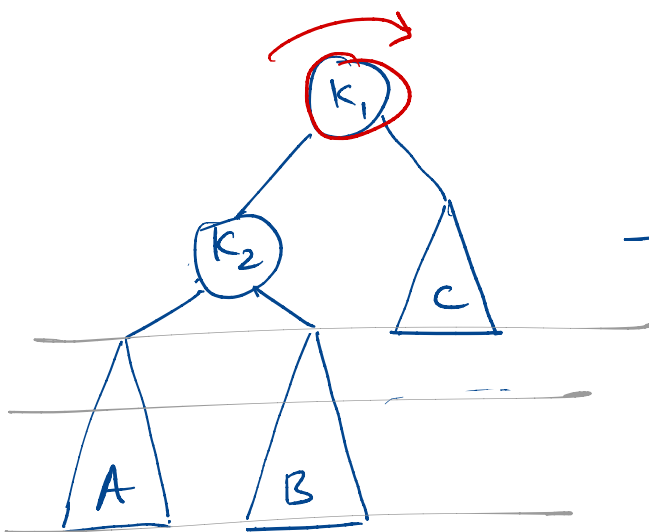
→ Two additional case beyond these

→ Offending subtree has a balance factor of 0

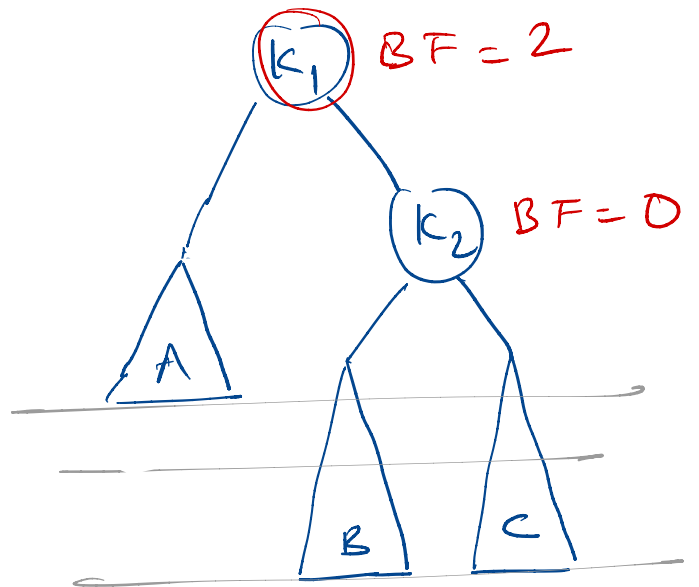
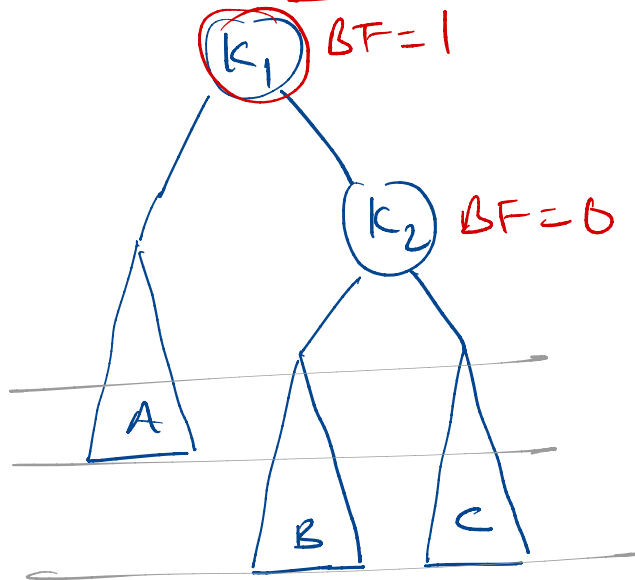


remove from C

Resolution: Perform a right rotation about K_1

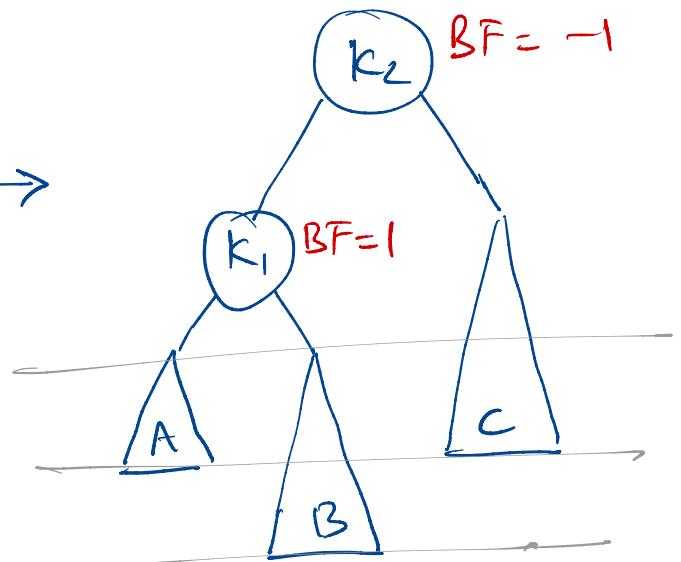
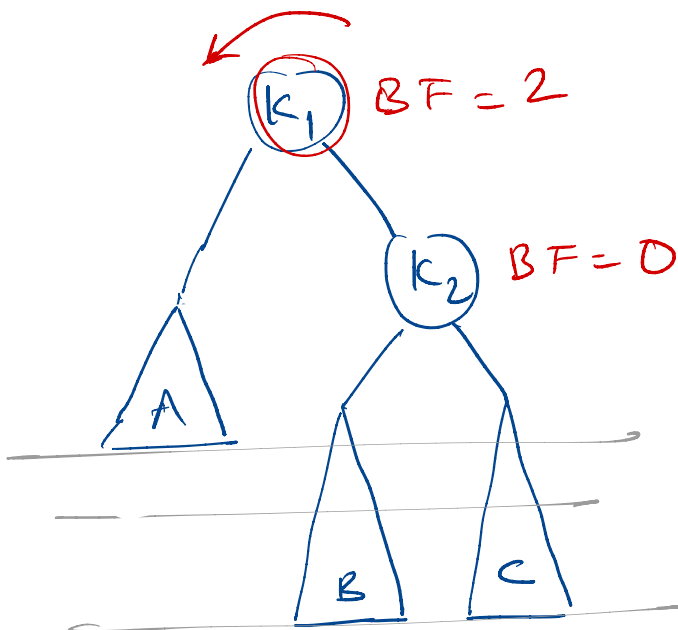


Cone 'G' R-both



Remove from A

Resolution: Rotate left about K_1



Implementing remove in a AVL tree :

↳ perform normal BST remove and fix imbalance at a node while going up

↳ Logic to detect the case for resolution :

• Current node $BF < -1 \Rightarrow$ Case LL, LR, or L-both

↳ look at current node's left child BF

fix with R rotation — left child $BF < 0 \Rightarrow$ Case LL

fix with LR rotation — left child $BF > 0 \Rightarrow$ Case LR

fix with R rotation — left child $BF = 0 \Rightarrow$ Case L-both

• Current node $BF > 1 \Rightarrow$ Case RR, RL or R-both

↳ look at current node's right child BF

fix with RL rotation — Right child $BF < 0 \Rightarrow$ Case RL

fix with L rotation — Right child $BF > 0 \Rightarrow$ Case RR

fix with L rotation — Right child $BF = 0 \Rightarrow$ Case R-both

Complexity : Adding $O(\lg n)$

removing $O(\lg n)$